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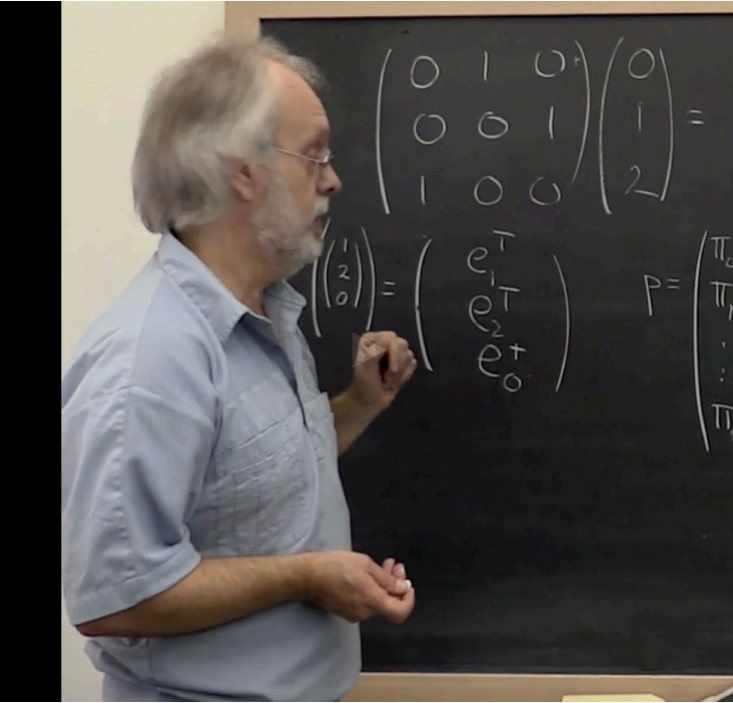
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the standard basis
vector e_1 transposed
and then e_2
transposed and then
 e_0
transposed. Now, we
know that if we
multiply this
times a vector, it picks
out the entry index
with 1, the entry
indexed with 2,
the entry indexed with
0. So standard basis
vectors can be
used to pick out
entries in a vector
when you take the dot
product. So the
other thing you notice
is that if we had only
had these indices,
that we would

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Reading assignment

1/1 point (graded)



Read Unit 5.3.2 of the notes. [\[LINK\]](#)

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✓ Correct (1/1 point)

Discussion

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Topic: Week 05 / 05.3.2

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	Cost of permutation matrix application?	3	▼
Suppose there is a matrix A with dimensions n by n and we wish to compute P...			
	Small typo in solution to HW 5.3.2.4.	2	▼

vehicle there is going to be something slightly simpler. So when we get to this point what we want is we want a permutation matrix, and we're going to call it P tilde, that given an integer, relative to this vector swaps the row that is relative to this row. So counting this row as being row 0, this integer tells us which row amongst these rows to swap. Okay? So that matrix actually has a very special form. That matrix P tilde of n , now let's just say

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1/1 point (graded)

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } x = \begin{pmatrix} \chi_0 \\ \vdots \\ \chi_{n-1} \end{pmatrix}.$$

$$P(p)x = \begin{pmatrix} \chi_0 \\ \vdots \\ \chi_{n-1} \end{pmatrix}$$

✓

$$P(p)x = \begin{pmatrix} \chi_{\pi_0} \\ \vdots \\ \chi_{\pi_{n-1}} \end{pmatrix}$$



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Homework 5.3.2.2.

1/1 point (graded)
Let

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } A = \begin{pmatrix} \tilde{a}_0^T \\ \vdots \\ \tilde{a}_{n-1}^T \end{pmatrix}.$$

Evaluate $P(p) A$.

☒ $P(p) A = \begin{pmatrix} \tilde{a}_{\pi_0}^T \\ \vdots \\ \tilde{a}_{\pi_{n-1}}^T \end{pmatrix}$

☐ $P(p) A = \begin{pmatrix} \tilde{a}_0^T \\ \vdots \\ \tilde{a}_{n-1}^T \end{pmatrix}$



Solution:

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

Homework 5.3.2.3.

1/1 point (graded)
Let

$$p = \begin{pmatrix} \pi_0 \\ \vdots \\ \pi_{n-1} \end{pmatrix} \text{ and } A = (a_0 \quad \cdots \quad a_{n-1}).$$

Evaluate $AP(p)^T$.

☒ $AP(p)^T = (a_{\pi_0} \quad \cdots \quad a_{\pi_{n-1}})$

☐ $AP(p)^T = (a_0 \quad \cdots \quad a_{n-1})$



Solution:

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

Homework 5.3.2.4.

1/1 point (graded)
Evaluate $P(p)P(p)^T$.

- ☒ $P(p)P(p)^T = I$
- ☐ $P(p)P(p)^T = P(p)$
- ☐ $P(p)P(p)^T = P(p)^T$



Solution:

For a complete solution, see [the notes](#).

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